



AutoTech Manufacturing Corp.

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IATF 16949 Certified | Engineering Division

BILL OF MATERIALS (BOM)

DOCUMENT NO.	REVISION
AT-BOM-4829	Rev 1.5

BOM INFORMATION		REFERENCE INFORMATION	
BOM Revision:	Rev 1.5	Customer PO:	PO-2025-15903
BOM Date:	October 10, 2025	Customer:	AutoTech Manufacturing Corp.
Status:	Approved	Project Code:	PRJ-AT-2025-Q4B
Effective Date:	October 10, 2025	Prepared By:	Technical Dept.
Supersedes:	Rev 1.4 (Sep 30, 2025)	PO Line Item:	Line 3 - 150 pcs
Document No.:	AT-BOM-4829	Supplier:	Precision Steel & Components Ltd. (SUP-PSC-2024-456)

MATERIAL IDENTIFICATION			
ID	TYPE/PART DESIGNATION	DIMENSIONS	MATERIAL GRADE
MAT-4829	Seamless Carbon Steel Pipe	NPS 2" (60.3 mm OD) x SCH 40 x 6000 mm	ASTM A106 Grade B
Material Overview: Hot-finished seamless carbon steel pipe suitable for high-pressure applications, structural supports, and mechanical systems. Manufactured to ASTM A106 Grade B specifications for enhanced strength and reliability.			

TECHNICAL SPECIFICATIONS			
Material Standard:	ASTM A106 Grade B / ASME SA-106 B	Pipe Type:	Seamless (no weld seam)
Nominal Pipe Size:	NPS 2" (DN 50)	Outside Diameter (OD):	60.3 mm (2.375 inches)
Schedule:	SCH 40 (Schedule 40)	Wall Thickness:	3.91 mm (nominal)
Inside Diameter (ID):	52.48 mm (nominal)	Length:	6000 mm (6 meters)
End Finish:	Plain end (PE), square cut	Surface Finish:	Black finish (as-rolled)
Length Tolerance:	+50 mm / -0 mm	Weight per Pipe:	22.3 kg (approx.)
Testing Required:	Hydrostatic test, mill test certificate (MTC) 3.1, dimensional check	OD Tolerance:	±0.4 mm (per ASTM A530)
Wall Tolerance:	-12.5 % of nominal	Dimensional Standard:	ASME B36.10M

CHEMICAL COMPOSITION (SPEC. LIMITS)		MECHANICAL PROPERTIES (MIN.)	
Element	Requirement (wt %)	Property	Requirement
Carbon (C)	0.30 max	Tensile Strength	415 MPa (60 ksi)
Manganese (Mn)	0.29 - 1.06	Yield Strength	240 MPa (35 ksi)
Phosphorus (P)	0.035 max	Elongation (long.)	Per ASTM A106 Table 4
Sulfur (S)	0.035 max	Hydrostatic Test	Per ASTM A106 / A530
Silicon (Si)	0.10 min	Bend / Flattening	Per ASTM A106 Sec. 11-12
Chromium (Cr)	0.40 max	Cr + Cu + Mo + Ni + V combined shall not exceed 1.00 %. Requirements per ASTM A106/A106M, Grade B.	
Copper (Cu)	0.40 max		
Molybdenum (Mo)	0.15 max		
Nickel (Ni)	0.40 max		
Vanadium (V)	0.08 max		



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QUALITY & INSPECTION PLAN				
#	Characteristic	Method / Reference	Frequency	Record
1	Chemical composition	MTC 3.1 review vs. ASTM A106	Each heat	MTC 3.1
2	Mechanical properties	Tensile per ASTM A370	Each lot	MTC 3.1
3	Hydrostatic test	Per ASTM A530, 5 s hold min.	Every pipe	MTC 3.1
4	OD & wall thickness	Micrometer / UT gauge, both ends	10 % of pipes	Inspection report
5	Length & end squareness	Tape & square	10 % of pipes	Inspection report
6	Seamless verification	Visual / mill route confirmation	Each heat	MTC 3.1

PACKAGING, MARKING & HANDLING	
Bundling:	Hexagonal bundles of 19 pcs, steel strapped at 3 points; wooden spacers
Protection:	Plastic end caps both ends; light mill varnish for transit
Marking:	Stenciled: manufacturer, ASTM A106 B, NPS 2 SCH 40, heat no., HYDRO, length
Bundle Tag:	Material ID MAT-4829, PO-2025-15903, pcs count, heat no., gross weight
Storage / Handling:	Store on dunnage off ground; cover to prevent rain exposure

ADDITIONAL SERVICES & FEES				
Fee (per supplier price list)	Applies	Charge basis	Quantity	Notes
Cutting Fee	No	-	-	Supplied at 6000 mm, plain end square cut as standard; no cutting
Cert Fee (MTC 3.1)	Yes	Per heat/lot	1 certificate	All 150 pipes supplied from a single heat
Hydrostatic Testing	Yes	Per pipe	150 pipes	Every pipe tested per ASTM A530; results stated on the MTC
Testing Fee	No	-	-	Not charged; hydrostatic testing above is the only testing charge for this item
Packaging (Standard / Export Grade)	No	-	-	Standard strapped bundles; no pallet or crate charge
Rush / Same-Day Processing	No	-	-	Standard lead time; no surcharge

Fee names and rates as listed in the Precision Steel & Components Ltd. Master Price List, effective October 01, 2025. Fees marked **Yes** are charged in the quantity shown and are **not included** in the PO unit price or PO total; they are invoiced as separate lines. Fees marked **No** are not charged for this item. No volume discount applies.

REVISION HISTORY				
Rev	Date	Description of Change	By	Appr.
Rev 1.3	Aug 21, 2025	Added ASME SA-106 B dual certification.	J. Okafor	K. Brennan
Rev 1.4	Sep 30, 2025	Length tolerance changed to +50 / -0 mm.	J. Okafor	K. Brennan
Rev 1.5	Oct 10, 2025	Linked to PO-2025-15903; hydrostatic test made mandatory for every pipe.	J. Okafor	K. Brennan

CONTACTS & APPROVALS			
Role	Name / Department	Signature	Date
Prepared	Technical Dept. - James Okafor	/s/ James Okafor	Oct 09, 2025
Checked	Laura Chen, Quality Engineer	/s/ Laura Chen	Oct 10, 2025
Approved By	Quality Assurance Department - Kevin Brennan, QA Manager	/s/ Kevin Brennan	Oct 10, 2025
Technical Contact	Thomas Richardson, Technical Sales Precision Steel & Components Ltd. (Supplier)	t.richardson@precisionsteel.com	+1 (412) 555-8912

Document Control: This BOM is controlled by AutoTech Manufacturing Corp. All pipes must be seamless and manufactured per ASTM A106 Grade B specifications.